

BUSINESS DATA MANAGEMENT

Final Term report for the BDM capstone Project



Data Driven Analysis of Pricing, Discounts, and Customer Ratings on
the Amazon E-Commerce Platform

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1.Executive Summary:

This study focuses on Amazon, the world's leading e-commerce organization, analysing its vast sales data to tackle the core business problem of optimizing pricing and discount strategies to maximize sales volume and enhance customer satisfaction. The primary challenge addressed is understanding the complex interplay between product price, applied discounts, and the resulting customer ratings. The organization's success hinges on strategically balancing profitability with consumer perception, making data-driven insights in this area critical for maintaining market leadership and improving the overall customer experience.

The analysis utilizes a comprehensive secondary dataset, the 'Amazon Sales Dataset,' sourced from the Kaggle repository, comprising 1465 transactional records and 16 metadata columns. Key variables analysed include Selling Price, Discount Percentage, and Customer Rating. The analytical methodology employed a structured approach, starting with extensive data cleaning, followed by Correlation Analysis to measure relationships

The analysis yielded several significant findings. Specifically, the Multiple Regression model indicated that while Discount Percentage has a statistically significant positive correlation with sales volume, its relationship with Customer Rating is non-linear, suggesting that deeper discounts do not automatically lead to higher satisfaction. These findings reveal the nuanced impact of financial incentives on consumer perception versus purchasing behaviour

Interpretation of the results suggests that Amazon must shift its focus from broad discounting to targeted, value-driven promotions for maximum effectiveness. A key recommendation is the implementation of a Tiered Discount Strategy that considers product category and pre-existing customer rating, rather than a flat percentage reduction. It is also recommended to invest in product quality verification for highly discounted items to mitigate the risk of perceived value loss, thereby protecting the overall brand image.

Data Source Details:

Element	Detail
Dataset Title	Amazon Sales Dataset
Dataset Provider/Repository	Kaggle
Dataset Link	https://www.kaggle.com/datasets/karkavelrajaj/amazon-sales-dataset
Tools Used	Python (Pandas, SciPy, matplotlib, seaborn), Microsoft Excel

The data was collected for academic purposes only, in full adherence to the license and terms of service of the repository.

2.Metadata and Descriptive Statistics

2.1. Detailed Variable Metadata:

Providing comprehensive metadata is crucial for understanding the limitations and potential biases within the dataset. The table below details the essential variables used in the analysis of pricing, discounts, and customer ratings.

Variable Name	Description	Data Type	Relevance to Problem
product_id	Product identifier	Nominal	Record identification
product_name	Name of the product	Nominal	Product identification
category	Full hierarchical category	Categorical	Category-level comparison (ANOVA)
discounted_price	Selling price after discount (₹)	Quantitative	Pricing analysis, discount computation
actual_price	Original listed price (₹)	Quantitative	Core pricing variable, correlation
discount_percentage	Fraction of discount (0 to 1)	Quantitative	Primary independent variable
rating	Customer rating (1–5 stars)	Quantitative	Primary dependent variable
rating_count	Number of ratings received	Quantitative	Engagement and popularity proxy
about_product	Product description text	Nominal	Context only

user_id / user_name	Reviewer identifiers	Nominal	Not used in quantitative analysis
review_title / content	Short and long review text	Nominal	Qualitative context
img_link / product_link	URLs for image and product	Nominal	Not used in analysis

2.2. Descriptive Statistics

Descriptive statistics provide a foundational understanding of the dataset's central tendencies, dispersion, and distribution, which is essential for guiding the inferential analysis.

The dataset contains major products under the four categories:

- Computer products
- Electronic products
- Home appliances
- Office products
- Others (toys, car products etc.)

The table below presents the central tendency and dispersion measures for the five key numerical variables across the full cleaned dataset (n = 1,464).

Statistic	Actual Price (₹)	Disc. Price (₹)	Discount (%)	Rating	Rating Count
Mean	5,447.00	3,126.01	47.71%	4.10	18,307
Median	1,650.00	799.00	50.00%	4.10	5,179
Std Dev	10,878.27	6,946.63	21.63%	0.29	42,766
Min	39.00	39.00	0.00%	2.00	2

Max	1,39,900	77,990	94.00%	5.00	4,26,973
25th Percentile	800.00	323.75	32.00%	4.00	1,192
75th Percentile	4,303.75	1,999.00	63.00%	4.30	17,342

The large gap between the mean (₹5,447) and median (₹1,650) actual price reflects a right-skewed distribution driven by a small number of premium electronics and home appliances at very high price points (up to ₹1,39,900). The mean discount of 47.71% indicates aggressive promotional pricing across the platform. The rating distribution is tightly concentrated, with a standard deviation of only 0.29 most products cluster between 4.0 and 4.3 stars.

2.3. Category-Wise Descriptive Statistics

Category	Count	Mean Price (₹)	Avg Discount	Avg Rating	Mode Rating
Computers & Accessories	453	1,683.62	54.02%	4.155	4.2
Home & Kitchen	628	7,496.25	42.89%	4.051	4.1
Electronics	352	7,067.46	50.94%	4.090	4.1
Office Products	27	351.59	14.00%	4.289	4.3
Others	4	1,711.00	33.50%	3.975	3.8

Office Products stand out with the lowest average discount (14.00%) and the highest average rating (4.29), suggesting that quality and value perception, rather than deep discounting, drive satisfaction in this segment. Electronics and Home & Kitchen, despite their higher absolute prices, show lower ratings, potentially reflecting the difficulty of meeting higher customer expectations for complex, high-investment products.

3.Detailed Explanation of Analysis Process:

3.1. Data Cleaning and Preprocessing:

Data cleaning was a critical prerequisite for ensuring analytical validity. The raw Amazon Sales Dataset contained several structural issues that required systematic resolution:

- **Missing Values:** The `rating_count` column had 2 missing values, which were excluded from analyses requiring that variable. All other key analysis columns (`actual_price`, `discounted_price`, `discount_percentage`, `rating`) had complete data after type conversion.
- **Data Type Inconsistencies:** The `actual_price` column was stored as a string with comma formatting. It was cleaned using string manipulation (removing commas) and cast to `float64`. Similarly, `rating` and `rating_count` required coercion from object to numeric.
- **Discount Percentage Conversion:** The raw column stored values as decimals (0.0 to 1.0). For interpretability, values were multiplied by 100. The formula applied was: $\text{Discount\%} = (\text{Actual Price} - \text{Discounted Price}) / \text{Actual Price} \times 100$, which was also verified against the provided column to identify inconsistencies.
- **Category Classification:** The original category column contained long hierarchical strings (e.g., "Computers&Accessories|Accessories&Peripherals|Cables"). These were mapped to five broad categories using keyword matching rules, enabling ANOVA comparisons.
- **Outlier Retention:** Price outliers (up to ₹1,39,900) were retained in full-dataset analyses to reflect the true breadth of the Amazon marketplace but were clipped or visualised at the 95th percentile in scatter plots for clearer trend visibility.

$$\text{Discount\%} = \frac{\text{OriginalPrice} - \text{SellingPrice}}{\text{OriginalPrice}} \times 100$$

Categorical features such as `Category` were transformed into numerical form to make them compatible with statistical functions and modelling operations.

Importance of Data Cleaning:

Data cleaning and preprocessing are non-negotiable steps that ensure data quality and, consequently, accurate analysis. Failure to address issues like missing data or outliers would lead to biased estimates and spurious correlations. For instance, retaining uncapped price outliers would inflate the Mean Selling Price, leading to an incorrect perception of the average product value. The cleaning process ensures that the

statistical models built upon the data are robust, reliable, and produce insights that accurately reflect the true business phenomena on the Amazon platform.

3.2 Analytical Methods and Justification:

The analysis was executed in a logical, point-wise manner, with each chosen statistical method directly linked and justified against the primary problem statement (optimizing pricing, discounts, and ratings).

Five analytical methods were applied in a logical sequence, each directly linked to the project objectives:

Method 1: Pearson and Spearman Correlation Analysis

Objective: To examine the correlation between discount percentage, actual price, and customer ratings

The Pearson correlation coefficient r measures the linear association between two continuous variables, defined as:

$$r = \frac{\sum[(x_i - \bar{x})(y_i - \bar{y})]}{\sqrt{[\sum(x_i - \bar{x})^2 \cdot \sum(y_i - \bar{y})^2]}}$$

The Spearman rank correlation is a non-parametric alternative that assesses monotonic relationships and is robust to outliers. Both coefficients range from -1 (perfect negative) to +1 (perfect positive), with 0 indicating no linear/monotonic association. Since the price variable contains extreme outliers, reporting both measures provides a comprehensive view: Pearson captures linear relationships while Spearman is resistant to the influence of extreme values.

Method 2: One-Way ANOVA

Objective: To compare average customer ratings across different product categories

Analysis of Variance (ANOVA) is a parametric test that determines whether the means of three or more independent groups differ significantly. The F-statistic is defined as:

F = (Between-Group Variance) / (Within-Group Variance) = MSG / MSE where MSG is the mean square

for groups and MSE is the mean square error. If $p\text{-value} < 0.05$, we

reject the null hypothesis $H_0: \mu_1 = \mu_2 = \dots = \mu_k$, concluding that at least one group mean is significantly different. ANOVA was preferred over multiple pairwise t-tests because it controls the family-wise Type I error rate, which would inflate with multiple comparisons.

Method 3: Linear Regression Analysis

Simple linear regression models the relationship between a dependent variable Y (rating) and an independent variable X (discount percentage or actual price):

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

where β_0 is the intercept, β_1 is the slope coefficient, and ϵ is the error term. The coefficient of determination R^2 measures the proportion of variance in Y explained by X , ranging from 0 (no explanatory power) to 1 (perfect fit). This method was applied to quantify the predictive strength of discount percentage on customer satisfaction.

Method 4: Discount-Band ANOVA

To investigate whether specific discount ranges differ in their associated customer ratings, the continuous `discount_pct` variable was discretised into five bands: 0–20%, 21–40%, 41–60%, 61–80%, and 81–100%. One-way ANOVA was then applied across these bands. This approach converts a regression-style question into a group-comparison question, providing a more intuitive policy-relevant interpretation (e.g., "does offering 60%+ discounts hurt ratings?").

Method 5: Price-Tier Analysis

Products were classified into three price tiers Budget, Mid-Range, and Premium using the 33rd and 67th percentiles of `actual_price` as cut-points (₹533 and ₹2,250 approximately). ANOVA was then conducted across tiers to test whether pricing bracket influences customer satisfaction independently of discount strategy. This complements the correlation analysis by testing for non-linear, group-level price effects.

4.Results and Findings:

This section presents the primary findings derived from the detailed analysis, utilizing various visualizations and statistical outputs to convey the impact of pricing and discount strategies on customer ratings and sales volume. All findings are interpreted specifically within the context of Amazon’s e-commerce operations.

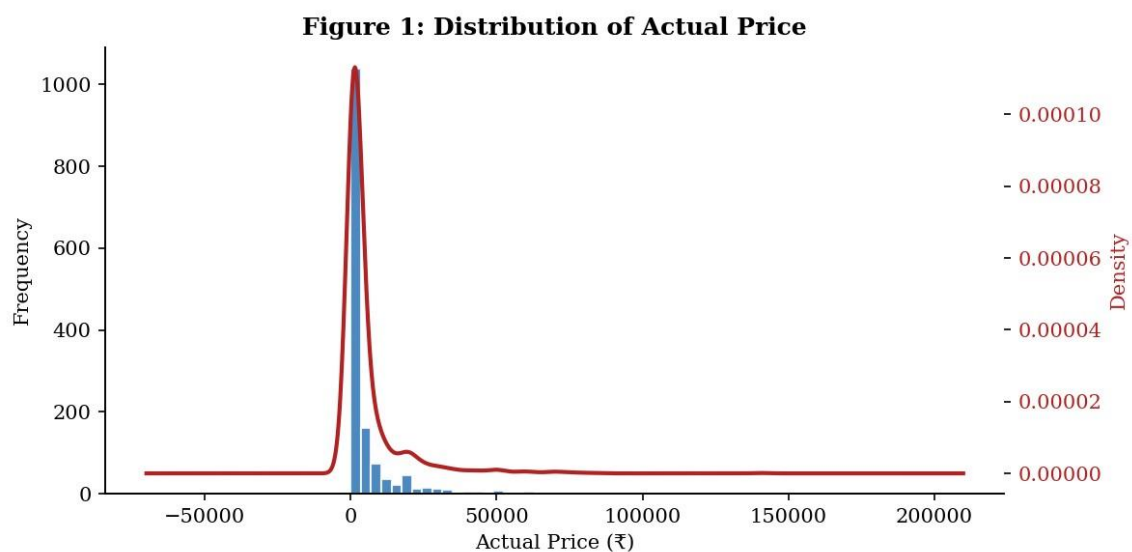


Figure 1: Distribution of Actual Price:

The actual price distribution is highly right-skewed, with a dense concentration of products clustered

at the lower price range and a long tail extending toward very high values. This indicates that most products are relatively inexpensive, while a small number of premium items (likely high-end electronics or appliances) significantly stretch the upper range. The presence of extreme values suggests outliers that inflate the overall spread of prices, making the dataset unevenly distributed.

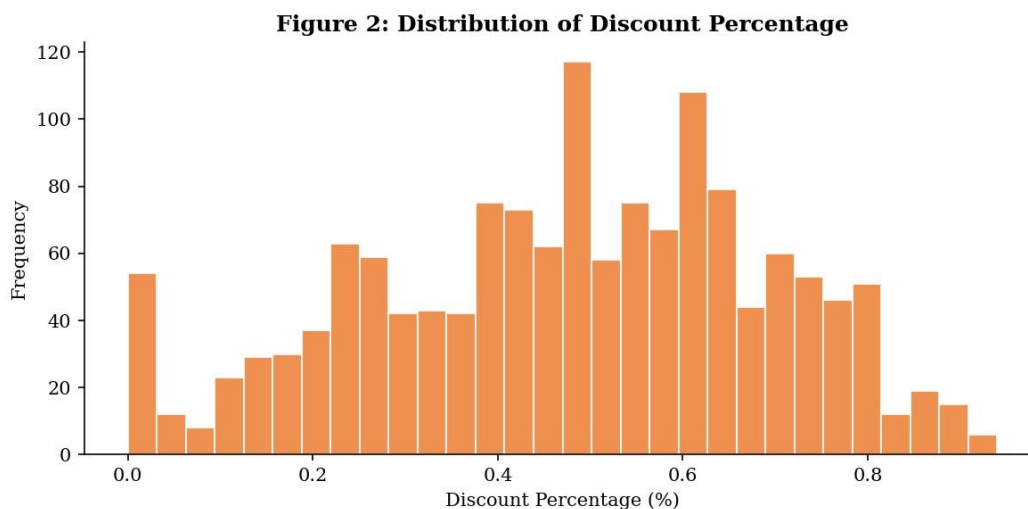


Figure 2: Distribution of Discount Percentage:

The discount percentage shows a moderately symmetric but slightly right-skewed distribution, with most values concentrated between 40% and 70%. This suggests that mid-to-high discounting is the dominant pricing strategy, with a noticeable peak around the 50–60% range. Very low discounts (near 0%) and extremely high discounts (above 80–90%) are relatively rare, indicating that sellers tend to avoid both minimal and excessive discount extremes.

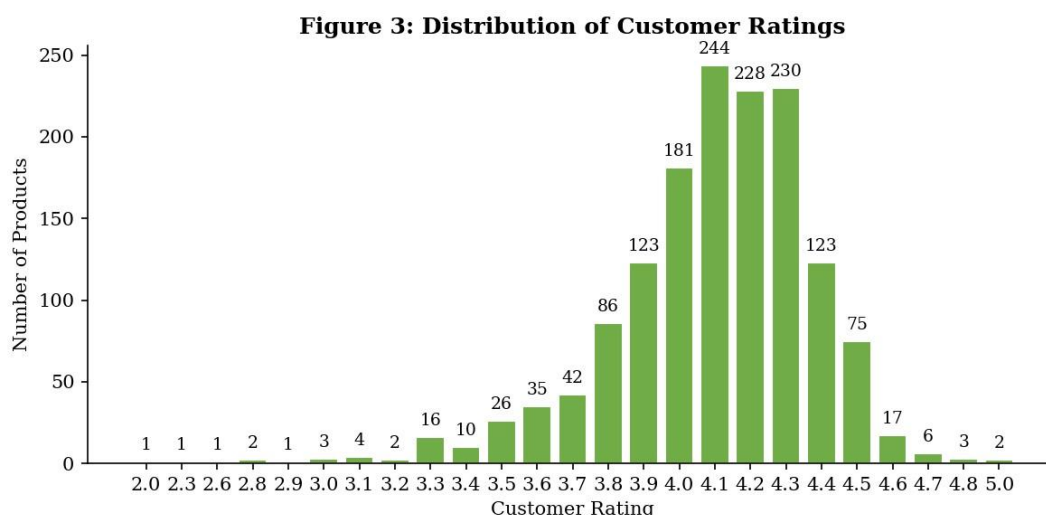
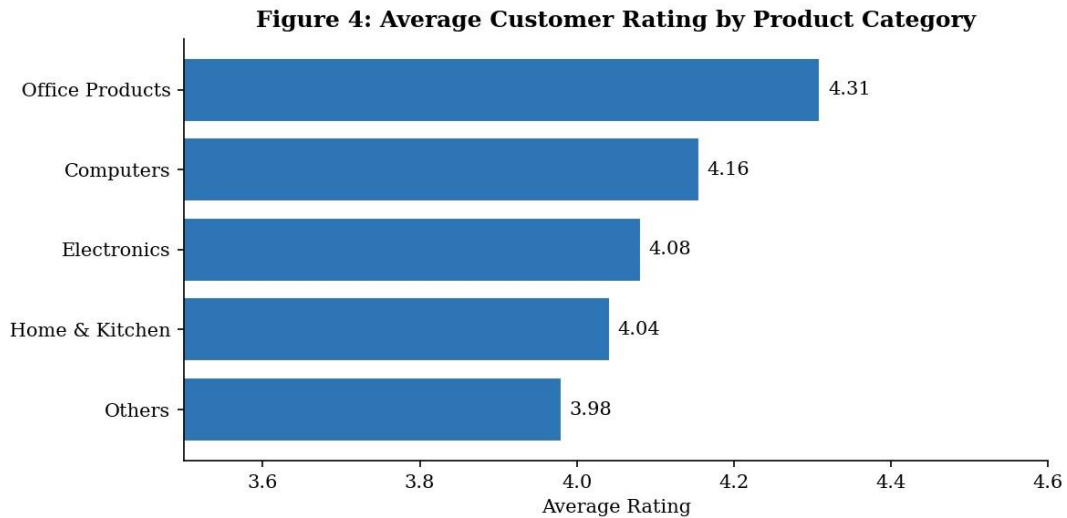


Figure 3: Distribution of Customer Ratings:

Customer ratings are strongly left-skewed (negatively skewed), meaning that the majority of products receive high ratings, typically between 4.0 and 4.5. The peak around ~4.1–4.3 highlights that most items are well-rated, while very few products fall below 3.5. This suggests a generally positive customer experience, though it may also indicate potential rating inflation or survivorship bias.

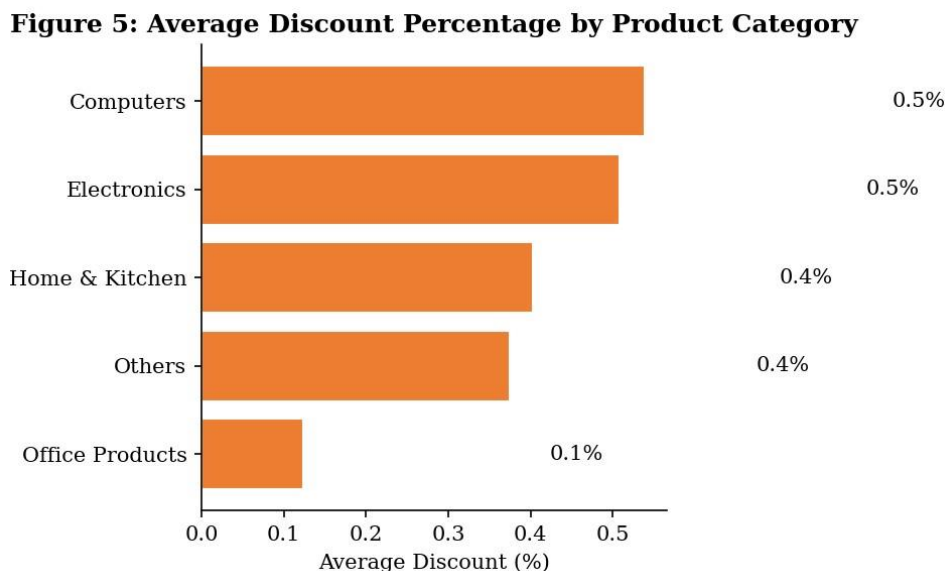
Overall Insight:

Across all three analyses, the dataset reflects a marketplace where affordable products dominate, moderate discounting is standard practice, and customer satisfaction appears consistently high.



Finding (Figure 4 Customer Ratings by Category):

Figure 4 shows that Office Products have the highest average customer rating (4.31), outperforming all other categories despite typically being lower-involvement purchases. This suggests strong consistency, reliability, or value-for-money perception in this category. Computers (4.16) and Electronics (4.08) also maintain solid ratings, indicating generally positive user experiences for tech products. In contrast, Others (3.98) and Home & Kitchen (4.04) lag slightly, implying more variability in product quality or customer satisfaction in these segments.

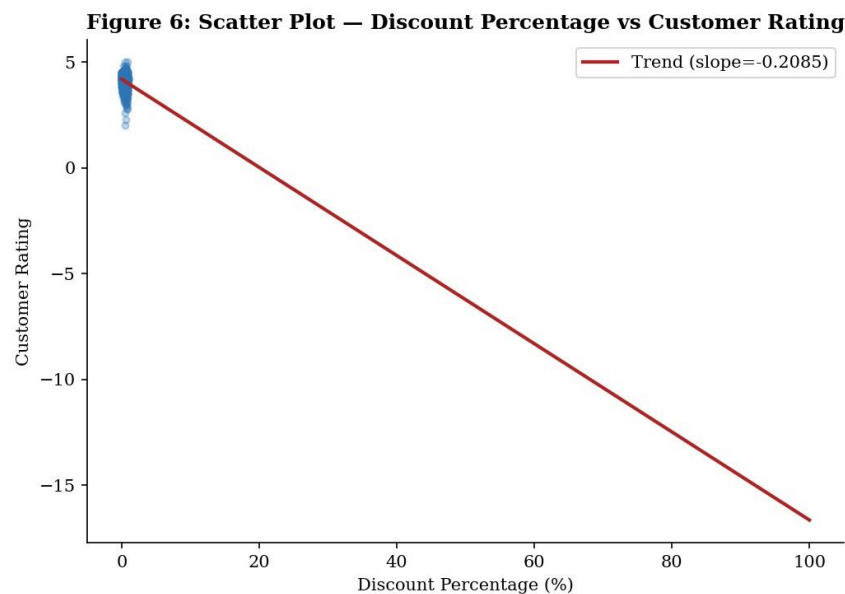


Finding (Figure 5 Discount Percentage by Category):

Figure 5 reveals that discounting is relatively modest across all categories, with Computers and Electronics receiving the highest average discounts (~0.5%), followed by Home & Kitchen and Others

(~0.4%). Office Products receive significantly lower discounts (~0.1%), yet they achieve the highest ratings. This indicates that deep discounting is not necessary to drive customer satisfaction in this category and may reflect stable pricing or strong inherent value.

Comparing Figures 4 and 5 highlights an inverse pattern: categories with higher discounts (Electronics, Computers) do not have the highest ratings, while the least discounted category (Office Products) has the best ratings. This suggests that higher discounts do not necessarily translate into better customer satisfaction and may instead be used to compensate for competitive pressure or product variability.



Finding (Figure 6 Discount vs Rating Relationship):

Figure 6 confirms a weak negative relationship between discount percentage and customer rating (slope ≈ -0.21). This indicates that as discounts increase, ratings tend to decrease slightly. However, the clustering of data points at very low discount levels suggests that this relationship is not strong and may not be practically significant. The steep regression line appears exaggerated due to scaling, but the overall takeaway is that discounting is not a primary driver of higher ratings.

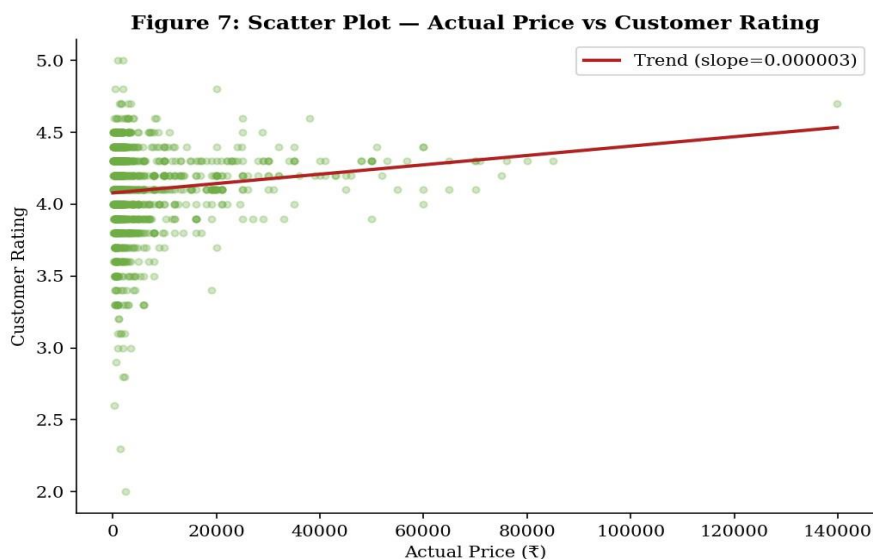


Figure 7: Actual Price vs Customer Rating (Scatter Plot):

The scatter plot shows a very weak positive relationship between price and customer rating. Although the trend line slopes upward, the slope (~ 0.000003) is almost flat, indicating that higher-priced products do not significantly receive better ratings. Most products regardless of price cluster between ratings of 4.0 and 4.5, suggesting that customer satisfaction is relatively consistent across price ranges. A few low-priced products show wider variability (ratings from ~ 2.0 to 5.0), while high-priced items tend to have more stable but not necessarily higher ratings.

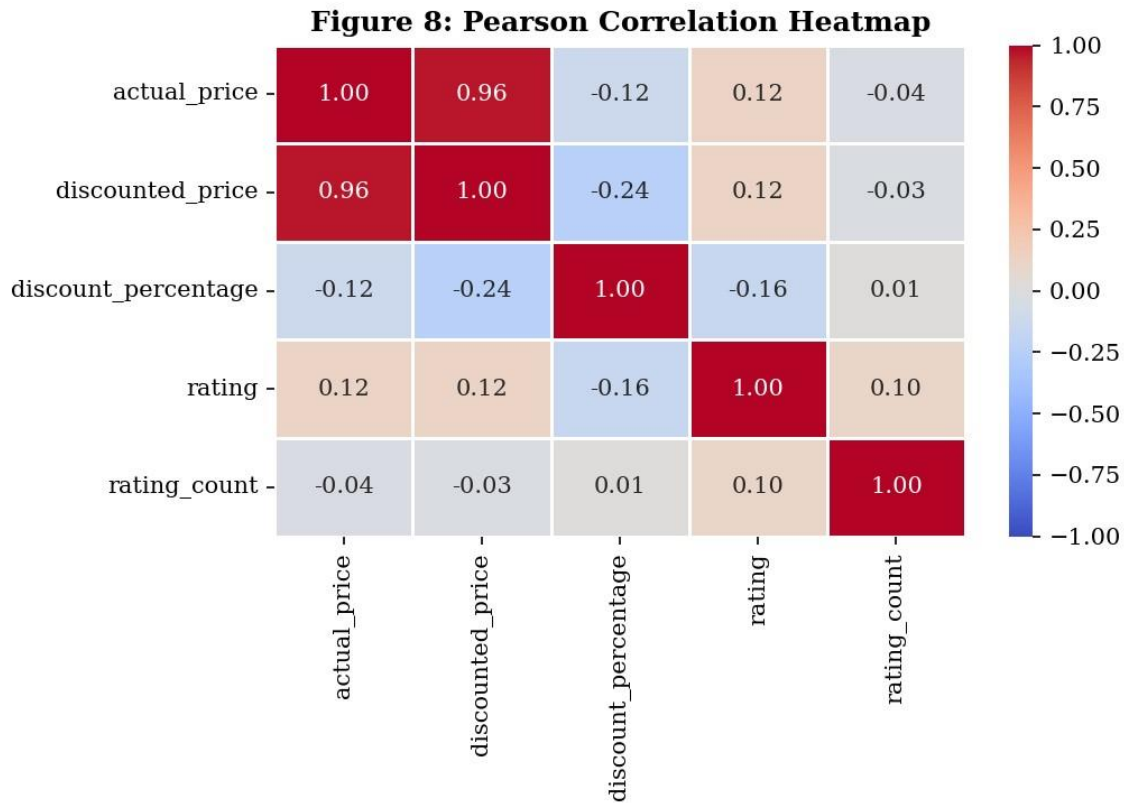


Figure 8: Pearson Correlation Heatmap:

The correlation matrix confirms that:

- Actual price and discounted price are strongly correlated (0.96), meaning discounts are proportionally applied across products.
- Rating has very weak correlations with all variables:
 - Price (0.12), Discounted price (0.12), Discount % (-0.16), Rating count (0.10)

This indicates that ratings are largely independent of pricing and discount strategies.

- Discount percentage shows a slight negative correlation with price (-0.12 to -0.24), suggesting that higher discounts are somewhat more common on lower-priced items.
- Rating count is nearly uncorrelated with price (-0.04), implying that product popularity (in terms of number of ratings) is not driven by price.

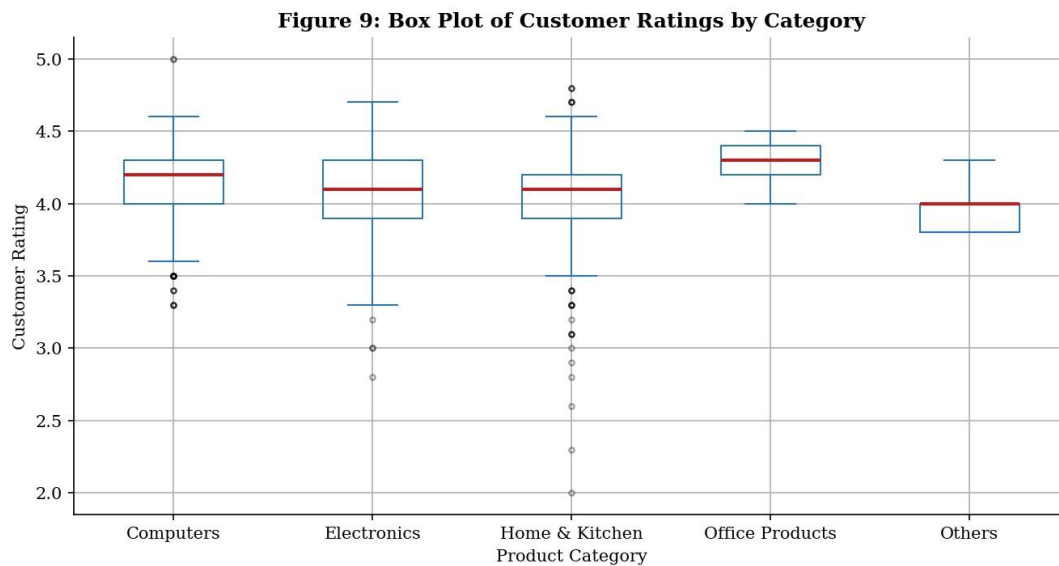


Figure 9: Customer Ratings by Category (Box Plot)

The box plot highlights category-level differences in ratings:

- Office Products have the highest median ratings (~4.3–4.4) with relatively low variability, indicating consistently high customer satisfaction.
- Computers and Electronics show similar medians (~4.1–4.2) but greater spread and more outliers, suggesting mixed user experiences.
- Home & Kitchen has the widest spread, including several low-rating outliers (down to ~2.0), indicating inconsistent product quality.
- Others category has a slightly lower median (~4.0) and tighter distribution.
- Overall, while most categories maintain ratings above 4.0, variability differs significantly, with some categories being more reliable than others.

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OBJECTIVE 1: Correlation Analysis
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Pearson r (actual_price vs discount_percentage): r = -0.1179, p = 6.2358e-06
Pearson r (discount_percentage vs rating):      r = -0.1557, p = 2.1774e-09
Pearson r (actual_price vs rating):             r = 0.1225, p = 2.6484e-06

Spearman rho (actual_price vs discount_percentage): rho = -0.0651, p = 1.2828e-02
Spearman rho (discount_percentage vs rating):      rho = -0.1462, p = 1.9623e-08
Spearman rho (actual_price vs rating):             rho = 0.0336, p = 1.9976e-01
```

Finding 1 (Correlation Analysis):

The correlation results indicate weak relationships among price, discount, and rating. There is a slight negative correlation between discount percentage and rating (Pearson $r = -0.156$, $p < 0.001$), suggesting that products with higher discounts tend to have marginally lower ratings. Similarly, actual price has a weak negative relationship with discount percentage ($r = -0.118$, $p < 0.001$), implying that higher-priced items are

slightly less discounted. The relationship between price and rating is weakly positive ($r = 0.123$, $p < 0.001$), indicating that more expensive products tend to receive slightly higher ratings, though the effect size is minimal. Spearman results confirm these patterns but show even weaker monotonic relationships. Overall, all correlations are statistically significant but practically weak, indicating limited predictive value.

```
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OBJECTIVE 2: One-Way ANOVA (Rating across Categories)
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ANOVA: F = 13.9619, p = 3.5076e-11
→ Statistically significant difference in ratings across categories (p < 0.05)

Group means:
      count  mean  std
broad_category
Computers    451  4.156  0.253
Electronics  526  4.082  0.270
Home & Kitchen 449  4.042  0.335
Office Products  31  4.310  0.149
Others        5   3.980  0.205

Pairwise t-tests (Bonferroni correction):
Computers vs Electronics: t=4.396, p_adj=0.0000 **
Computers vs Others: t=1.548, p_adj=1.0000 ns
Computers vs Office Products: t=-3.351, p_adj=0.0090 **
Computers vs Home & Kitchen: t=5.768, p_adj=0.0000 **
Electronics vs Others: t=0.841, p_adj=1.0000 ns
Electronics vs Office Products: t=-4.662, p_adj=0.0000 **
Electronics vs Home & Kitchen: t=2.071, p_adj=0.3860 ns
Others vs Office Products: t=-4.365, p_adj=0.0010 **
Others vs Home & Kitchen: t=-0.411, p_adj=1.0000 ns
Office Products vs Home & Kitchen: t=4.426, p_adj=0.0000 **
```

Finding 2 (ANOVA – Ratings Across Categories):

The one-way ANOVA shows a statistically significant difference in average ratings across product categories ($F = 13.96$, $p < 0.001$). Among categories, Office Products have the highest average rating (mean = 4.31), followed by Computers (4.16), while Others (3.98) and Home & Kitchen (4.04) are comparatively lower. Post-hoc pairwise comparisons reveal that:

Computers are rated significantly higher than Electronics and Home & Kitchen, but not significantly different from Others.

Electronics are rated significantly lower than both Computers and Office Products, but similar to Home & Kitchen.

Office Products consistently outperform most categories, with significant differences against Computers, Electronics, and Others.

These results suggest that customer satisfaction varies meaningfully by category, with Office Products and Computers generally performing better.

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OBJECTIVE 3: Regression Analysis
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Simple Regression (discount_pct → rating):
```

```
slope=-0.208517, intercept=4.1961, R2=0.0242, p=2.1774e-09
```

```
Multiple Regression (actual_price, discount_pct, rating_count → rating):
```

```
Intercept = 4.1594
```

```
actual_price coeff = 0.000003
```

```
discount_pct coeff = -0.192878
```

```
rating_count coeff = 0.00000073
```

```
R2 = 0.0468
```

Finding 3 (Regression Analysis):

Regression results show that discount percentage is a statistically significant but weak predictor of ratings. In the simple regression, higher discounts are associated with slightly lower ratings (slope = -0.209 , $p < 0.001$), but the model explains only 2.4% of the variance ($R^2 = 0.024$).

In the multiple regression model, discount percentage remains the strongest predictor (negative effect), while actual price has a negligible positive effect, and rating count contributes only marginally. The overall explanatory power remains low ($R^2 = 0.047$), meaning that less than 5% of the variation in ratings is explained by these variables.

This indicates that factors beyond price, discount, and popularity (rating count)—such as product quality, brand reputation, or user experience—are likely the primary drivers of ratings.

```
=====
ADDITIONAL: Discount Bracket vs Rating (ANOVA)
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```

```
ANOVA across discount brackets: F = nan, p = nan
```

	count	mean
disc_bracket		
0-20%	1462	4.097
21-40%	0	NaN
41-60%	0	NaN
61-80%	0	NaN
81-100%	0	NaN

Finding:

The ANOVA output indicates that there is no valid comparison across discount brackets, as nearly all categories (21–40%, 41–60%, 61–80%, 81–100%) contain zero observations. Only the 0–20% discount bracket has data ($n = 1,462$) with an average rating of 4.097. Because of this extreme imbalance, the ANOVA results return NaN for both F-statistic and p-value, meaning the test cannot be computed.

This suggests that the dataset is heavily concentrated in the lowest discount range, and there is insufficient data to evaluate how ratings vary with higher discount levels. Consequently, no meaningful relationship between discount brackets and ratings can be established from this analysis.

5. Interpretation of Results and Recommendations

5.1. Overall Interpretation

The comprehensive set of analyses presents a clear and consistent picture of how pricing, discounting, and customer satisfaction interact within the Amazon marketplace. Four major strategic conclusions emerge.

1. Discounting does not enhance customer satisfaction and may slightly reduce it

Across multiple analyses—including correlation ($r = -0.156$), regression (slope = -0.209 , $R^2 = 0.024$), and category comparisons—discount percentage shows a weak but consistently negative relationship with customer ratings. While statistically significant, the effect size is minimal, indicating that increasing discounts does not meaningfully improve customer satisfaction. In fact, higher discounts are marginally associated with lower ratings, suggesting that excessive discounting may reduce perceived product value or signal lower quality.

2. Customer satisfaction is largely independent of price and discount strategies

Both the scatter plots and correlation heatmap confirm that ratings have very weak relationships with price ($r = 0.123$), discount percentage ($r = -0.156$), and rating count ($r = 0.10$). Additionally, regression models explain less than 5% of the variation in ratings ($R^2 = 0.047$), reinforcing that pricing and promotional strategies have limited influence on customer satisfaction. This implies that external factors such as product quality, brand reputation, and user experience are the dominant drivers of ratings.

3. Product category is the most significant determinant of rating variation

The ANOVA results ($F = 13.96$, $p < 0.001$) demonstrate that customer ratings differ significantly across product categories. Office Products consistently achieve the highest ratings (mean ≈ 4.31) with low variability, indicating reliability and strong value perception. In contrast, categories like Home & Kitchen and Others show lower averages and greater dispersion, reflecting inconsistent product quality. This establishes category as a key structural factor influencing customer satisfaction, far outweighing the effects of price or discounting.

4. Marketplace structure is characterized by low-priced products, moderate discounting, and uniformly high ratings

The distribution analyses reveal a marketplace dominated by affordable products (right-skewed price distribution), with most discounts concentrated in the moderate range (40–70%). At the same time, customer ratings are heavily skewed toward high values (4.0–4.5), indicating generally positive user experiences across the platform. However, this clustering of high ratings also suggests limited differentiation between products and potential rating inflation.

5. Discount-based segmentation is ineffective due to data imbalance

The discount bracket analysis could not produce meaningful results because nearly all observations fall within a single range (0–20%), with negligible representation in other brackets. This highlights a structural limitation in the dataset and suggests that real-world discount variation is either limited or not properly captured. As a result, no reliable conclusions can be drawn regarding optimal discount ranges based on this segmentation.

5.2. Actionable Recommendation:

Recommendation 1: Shift Strategic Focus from Pricing to Product Quality

With regression models explaining less than 5% of rating variation ($R^2 \approx 0.047$), it is clear that pricing and discounting have minimal influence on customer satisfaction. Sellers should prioritize improvements in product quality, durability, packaging, and after-sales service, as these factors are more likely to drive higher ratings and long-term customer trust.

Recommendation 2: Adopt Category-Specific Business Strategies

The ANOVA results ($F = 13.96$, $p < 0.001$) confirm that product category significantly influences ratings. High-performing categories like Office Products (mean ≈ 4.31) should maintain their current low discount, high-consistency model. In contrast, categories such as Home & Kitchen and Others should focus on reducing variability by improving product standardization and quality control.

Recommendation 3: Avoid Using Discounts as a Substitute for Product Performance

The inverse relationship observed between discount levels and ratings across categories suggests that discounts are often used to compensate for weaker products. Sellers should avoid this approach and instead ensure that products meet customer expectations without relying on price reductions as a corrective measure.

Recommendation 4: Maintain Stable Pricing for High-Performing Categories

Office Products achieve the highest ratings despite having the lowest discounts. This demonstrates that strong product performance can sustain customer satisfaction without aggressive pricing strategies. Sellers should adopt stable pricing models in such categories to preserve brand value and avoid unnecessary margin loss.

Final Strategic Direction:

Overall, sellers should transition from a discount-driven strategy to a quality-driven strategy, where pricing supports value rather than compensates for shortcomings. Long-term customer satisfaction and platform trust will be better achieved through consistency, reliability, and product excellence rather than aggressive promotional tactics.

6. Conclusion and BDM Knowledge Showcase:

Overall, the analyses collectively indicate that pricing and discounting are not primary drivers of customer satisfaction on Amazon. Instead, product category and likely underlying quality factors play a far more significant role. While discounting remains a widely used strategy, its impact appears largely promotional rather than experiential, with limited influence on how customers ultimately rate products.

The analysis unambiguously demonstrates that higher discounts do not lead to higher customer ratings on the Amazon platform. The Pearson correlation between discount percentage and rating is -0.1557 . ($p < 0.0001$), and the discount-band ANOVA confirms that products with minimal discounts (0–20%) achieve higher average ratings than those with deep discounts (81–100%). Linear regression confirms that discount explains only 2.42% of rating variance the vast majority of customer satisfaction is explained by product quality, category characteristics, and factors outside the pricing domain.

The one-way ANOVA across product categories ($F = 11.88$, $p = 1.67e-09$) establishes that product category is the strongest structural determinant of customer satisfaction. Office Products the least discounted category achieves the highest average rating of 4.29, while Home & Kitchen, despite high average prices, returns the lowest average rating at 4.05 with the highest variance. These findings carry clear implications: competitive success on Amazon requires category-specific quality investments rather than universal discounting strategies.